America. Rock below South Africa was suddenly pulling more strongly; rock below North America was suddenly pulling more weakly! These changes must have been produced by some sort of bulk field that passes through our brane and controls *G*, he reasoned.

Bulk fields are not *just* the key to gravitational anomalies on Earth, Professor Brand believes (in my extrapolation). Bulk fields also play two other crucial roles: They hold the wormhole open, and they protect our universe from destruction.

Holding the Wormhole Open

The wormhole that connects our solar system to Gargantua's neighborhood, if left to its own devices, will pinch off (Figure 25.5). Our connection to Gargantua will be severed. This is the unequivocal conclusion of Einstein's relativistic laws (Chapter 14).

If there is *no* bulk, then the only way to hold the wormhole open is to thread it with exotic matter that repels gravitationally (Chapter 14). The dark energy that may accelerate our universe's expansion (Chapter 24) is probably not repulsive enough. In fact, it seems likely, in 2014, that the laws of quantum physics prevent even an exceedingly advanced civilization from ever collecting enough exotic matter to hold the wormhole open. And I imagine this conclusion is even more certain in Professor Brand's era.

But there is an alternative, the Professor realizes in my extrapolation of the movie's story. Bulk fields may do the job. They may hold the wormhole open. And since the Professor thinks the wormhole has been constructed and placed near Saturn by bulk beings, bulk fields holding it open seem natural to him.